UAV OPERATION

BRICS FUTURE Hi-Tech Skill Championship – Ekaterinburg, Russia

Presented by

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MODULE A

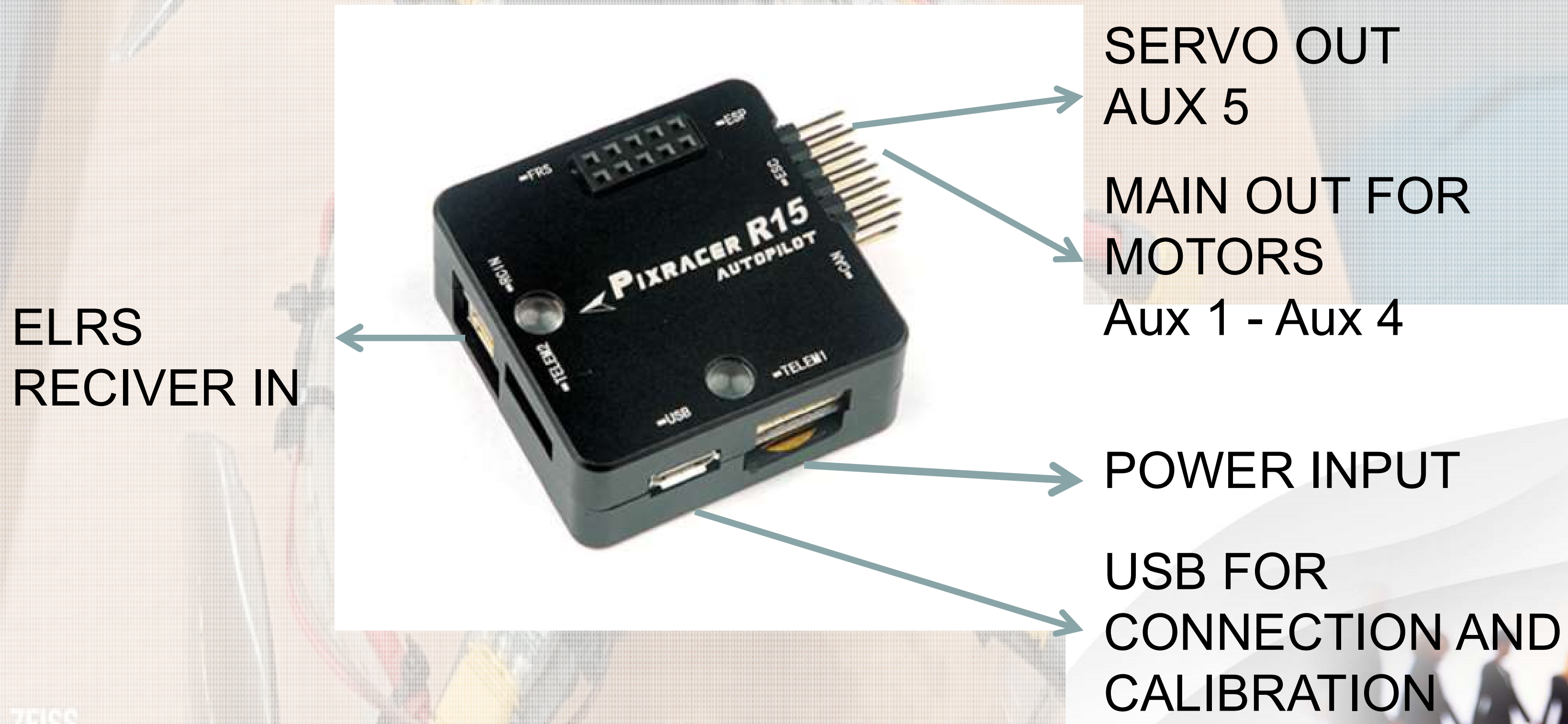
- Farmers face difficulties in transporting produce and materials across large fields.
- Manual transport is time-consuming and labor-intensive.
- Objective: Design and development of a cargo transportation unit for agriculture

Component Details

- Flight Controller: Pixracer
- Receiver: ELRS
- Transmitter: Radiomaster TX12 ELRS
- Motors: 2300KV BLDC (x4)
- Propellers: 5-inch Triblade
- Servo: 5V Metal Gear Servo
- Gripper: 3D Printed

Electrical Schematic

Electrical schematic diagram(showing connections for FC, motors, ESCs, servo, and power).



Working Principle

- •Drone hovers over cargo location.
- Servo gripper closes to pick up payload via AUX channel 5 control.
- Drone navigates to drop point.
- Servo opens to release payload.
- Demonstrates remote-controlled precision and stability.

Flight Controller Setup – Pixracer

- Configured through Mission Planner.
- AUX 5 output assigned to servo channel.
- Servo end-points calibrated.
- ELRS Receiver paired with Radiomaster TX12 for reliable signal control.

Servo Gripper Mechanism

- 3D printed mechanical hook attached to drone base.
- 5V Metal Gear Servo provides actuation.
- Controlled via transmitter switch mapped to Channel 5.
- Lightweight and efficient design to handle moderate payloads.

MODULE B

- The powerlines are huge source of electricity us so for the maintance of the powerlines we are using drones for it

- Objective : Maintance of powerlines using UAV

Project Objective

- Design and build a UAV for power line inspection and light maintenance tasks.
- Integrate a servo-controlled manipulator for simple mechanical repairs or cleaning.
- Optimize flight stability using Pixracer FC and ELRS control system.
- Enable safe remote operation near high-voltage environments.

Component Details

- Flight Controller: Pixracer
- Receiver: ELRS
- Transmitter: Radiomaster TX12 ELRS
- Motors: 2300KV BLDC (x4)
- Propellers: 5-inch Triblade
- Servo: 5V Metal Gear Servo
- Drone Arm: 3D Printed

Servo Arm Mechanism

- 3D-printed lightweight servo arm mounted under UAV.
- Controlled via transmitter switch mapped to Channel 5.
- Designed for minor maintenance like brushing insulators or placing sensors.
- Metal gear servo provides precision and strength.

Demonstration Summary

- Demonstration includes lifting and dropping a payload within the farm field.
- Showcases control, stability, and payload handling.
- Smooth servo operation and safe Landing on the powerlines with removal of obstacles are demonstrated.

Safety & Maintenance

- • Pre-flight check: Verify all connections and propellers.
- • Maintain 10-meter safety zone during operation.
- • Regular calibration of flight controller.
- • Inspect gripper mechanism before each flight.

Documentation Overview

- User manual for operation and maintenance.
- 3D model of the servo Arm
- Wiring schematic and assembly steps documented.
- Test results recorded.
- Data to be used for further system optimization.

Future Development

- Integrate GPS for Position Hold.
- Enable autonomous waypoint missions.
- Develop lightweight frame for improved efficiency.
- Add telemetry for real-time monitoring.
- Increase payload capacity.

Conclusion

- The cargo transport drone provides a practical, efficient, and scalable solution for farm logistics.
- The project demonstrates the integration of mechatronic systems with UAV technology for Powerline Inspection.
- Further advancements in automation and GPS control will enhance operational reliability.

REFERENCES

- https://px4.io/
- https://qgroundcontrol.com/
- https://github.com/mavlink/qgroundcontrol
- https://github.com/ArduPilot/ardupilot

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- https://github.com/sureshkumarbalasubramanianin/-BRICS-Hi---Tech-Future-Skill-Competetion_UAV-Operation_Team-INDIA